

CE 561: The science of climate change

Course Code	CE 561
Course Name	The science of climate change
L-T-P-C	3-0-0-3
Prerequisites	IC 230 (Environmental Science) or equivalent
Intended for	3rd and 4th year B.Tech., PG
Distribution	Elective
Approval	recommended by the 38th BoA (Jan 2021)

Course Contents

- **Earth system concepts and early climate:** State of a system and couplings; negative and positive feedback loops; system response to feedbacks; stable and unstable equilibrium states; application of these concepts to the Daisyworld climate system; formation and composition of the early atmosphere, and the effect of early life; the Great Oxidation Event and the ozone shield; early climate and its evolution over time. [5 Lectures]
- **Basics of global climate:** Components of the climate system, timescales and the parameterization problem; basics of energy balance and radiative forcing; atmospheric and oceanic circulation and their role in energy redistribution; conservation of momentum, equations of state, temperature and continuity equations, moist processes, wave processes; natural climate variability - El Nino Southern Oscillation (ENSO) dynamics, ENSO teleconnection and prediction. [7 Lectures]
- **Long-term climate regulation and proxies:** The Faint Young Sun paradox - a CO₂ and CH₄-rich early atmosphere; geological indicators of paleoclimate; isotopic proxies and temperature reconstruction; long-term glacial record and evidence of past glaciations; low-latitude glaciation and the Snowball Earth; banded-iron formations and cap carbonates; Pleistocene glaciations and the Milankovitch Cycles; climate record from the Dome C and Vostok ice cores - evidence of anthropogenic influence. [6 Lectures]
- **The greenhouse effect and climate feedbacks:** Global energy balance model with a 1-layer atmosphere - IR emissions and temperature; magnitude of the greenhouse effect; global warming potential; climate feedbacks and climate sensitivity - water vapour, snow/ice, cloud, lapse rate and stratospheric cooling feedbacks; climate response time and sensitivity in transient climate change - doubling of CO₂, ocean warming. [5 Lectures]
- **Modern climate and human influence:** Climate change during the Holocene and the industrial era; the carbon, nitrogen and water cycles and their perturbations, carbon emission intensity, estimating emissions from the Kaya Identity, CO₂-equivalents; radiative forcing of other greenhouse gases (CH₄, N₂O, halocarbons) and aerosols since industrialization. [7 Lectures]
- **Effects of climate change in the polar regions and the Himalayas:** Effects of climate change on the cryosphere; temperature and precipitation trends in the poles and the Himalayas under a changing climate; aerosol deposition, snow cover change and glacier melting; effects on stream flow and water resources; the National Action Plan on Climate Change and the National Mission for Sustaining the Himalayan Ecosystem; resilience and adaptation of Himalayan communities to climate change. [5 Lectures]
- **Climate projections for the future and adaptation/mitigation strategies:** Emission paths and scenarios; globally-averaged response to warming scenarios; coupled climate models, projections and multi-model ensemble averages; mitigation strategies - climate-smart agriculture, energy efficiency, GHG accounting, renewable energy, sustainable urban systems. [7 Lectures]

Text Books

1. J. David Neelin, **Climate Change and Climate Modeling**, 4th Edition, Cambridge University Press, 2015.
2. Lee R. Kump, James F. Kasting, Robert G. Crane, **The Earth System**, 3rd Edition, Prentice Hall., 2010

References

1. Eckhart Ehlers, Thomas Krafft, **Earth System Science in the Anthropocene**, Springer, 2005.
2. Barbara J. Finlayson-Pitts and James N. Pitts Jr, **Chemistry of the Upper and Lower Atmosphere**, 2nd Edition, Academic Press, 2000.
3. John Marshall and R. Alan Plumb, **Atmosphere, Ocean and Climate Dynamics**, Academic Press, 2007.
4. Relevant IPCC reports: AR5 Climate Change 2013 - The Physical Science Basis; Global Warming of 1.5 C; The Ocean and Cryosphere in a Changing Climate.
5. Research articles will be advised as required.